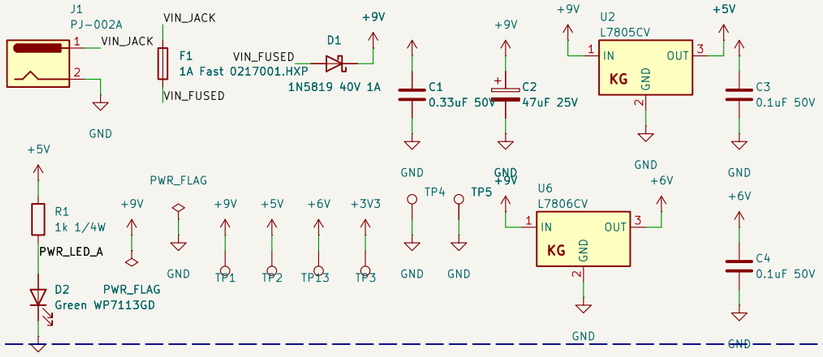
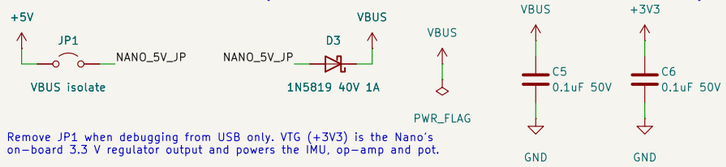
POWER: 9–12 V IN, FUSE, REVERSE PROTECTION, L7805 (5 V), L7806 (6 V MOTOR)

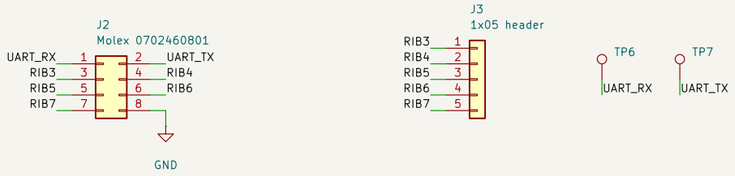


MCU POWER: 5 V TO NANO VBUS (USB BACK-FEED PROTECTION + ISOLATION JUMPER)



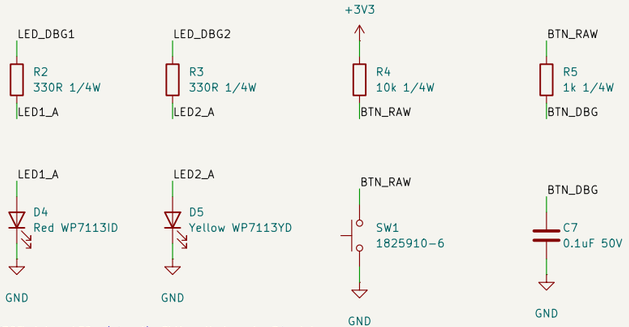
Remove JP1 when debugging from USB only. VTG (+3V3) is the Nano's on-board 3.3 V regulator output and powers the IMU, op-amp and pot.

TEAM RIBBON: 2x4 IDC TO HUB CONNECTOR 3 (PIN 8 = GND)



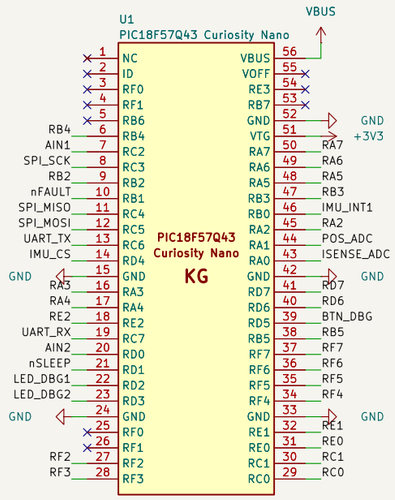
Pin 1 UART_RX: actuator B command, hub -> RC7 (U1RX).
Pin 2 UART_TX: IMU jerk/impulse + status, RC6 (U1TX) -> hub.
Pins 3–7 unused by the team; broken out on J3 for jumpering.

DEBUG: LEDS, HARDWARE-DEBOUNCED BUTTON



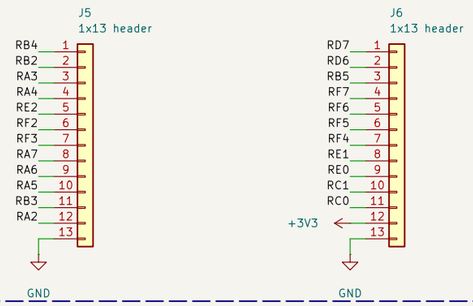
RD2/RD3 debug LEDs (~4 mA). SW1 pulled up by R4, debounced by R5/C7 (tau ~0.1 ms + 1 ms release via R4) into RD5. Nano also has on-board LED0 (RF3) and SW0 (RB4), both left free on J5/J6.

MICROCONTROLLER: PIC18F57Q43 CURIOSITY NANO

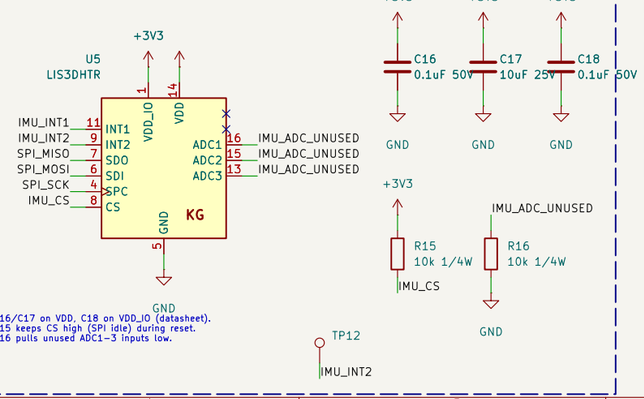


Not connected: RF0/RF1 (Nano CDC UART), RB6/RB7 (ICSP), RE3 (MCLR), VOFF (leave open = target powered), ID.

SPARE GPIO HEADERS (DEBUG / EXPANSION)

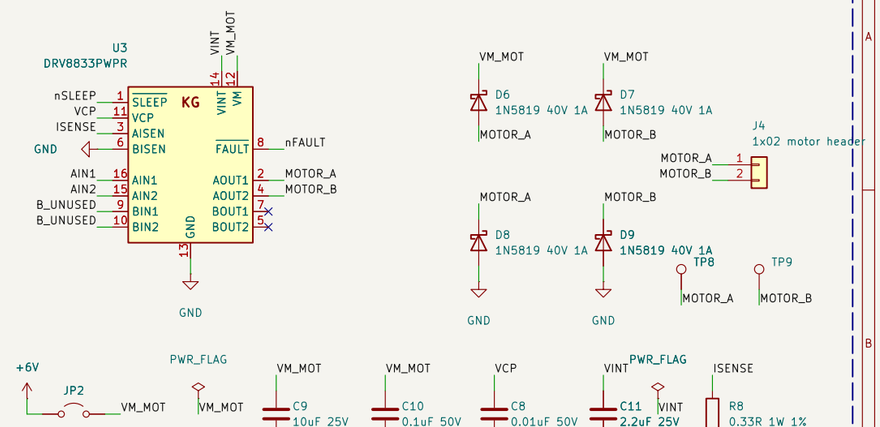


IMU: LIS3DHTR 3-AXIS ACCELEROMETER (SPI1) -> JERK / IMPULSE



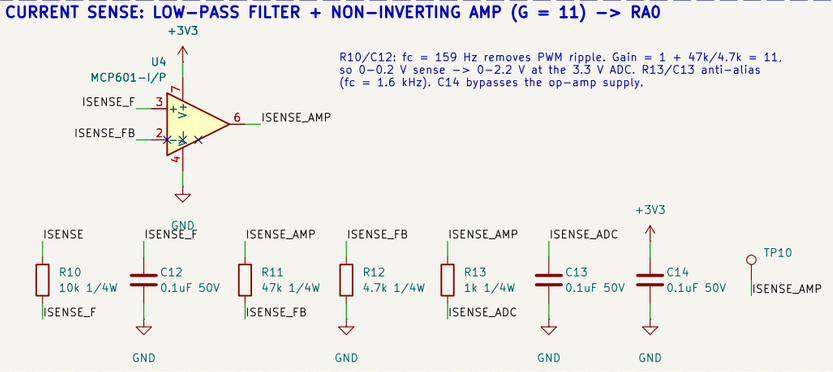
C16/C17 on VDD, C18 on VDD_IO (datasheet).
R15 keeps CS high (SPI idle) during reset.
R16 pulls unused ADC1–3 inputs low.

ACTUATOR B: DRV8833 H-BRIDGE -> N20 LEAD-SCREW GEARMOTOR



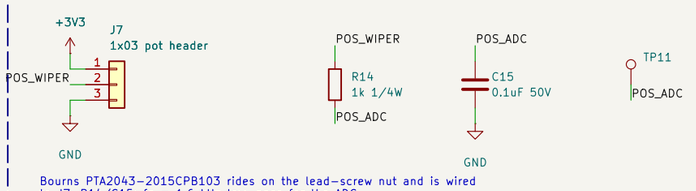
R8 sets the chopping limit: 0.2 V / 0.33 R = 0.61 A (adapter-safe).
R6 holds the driver asleep during MCU reset; R7 pulls unused bridge B inputs low. D6–D9 clamp motor back-EMF to VM/GND.
JP2 isolates the 6 V motor rail (U6) for debugging.

CURRENT SENSE: LOW-PASS FILTER + NON-INVERTING AMP (G = 11) -> RA0



R10/C12: fc = 159 Hz removes PWM ripple. Gain = 1 + 47k/4.7k = 11, so 0–0.2 V sense -> 0–2.2 V at the 3.3 V ADC. R13/C13 anti-alias (fc = 1.6 kHz). C14 bypasses the op-amp supply.

POSITION FEEDBACK: SLIDE POT (OFF-BOARD) -> RC FILTER -> RA1



Bourns PTA2043-2015CPB103 rides on the lead-screw nut and is wired to J7. R14/C15: fc = 1.6 kHz low-pass for the ADC.

RAS 304 Individual Schematic Design — Kaushik Gunasekaren (KG)
Custom KG symbols: Curiosity Nano, L7805CV, DRV8833, LIS3DH, MCP601
Spoke board on hub connector 3 of the Modular Spider Leg
Team 102: Modular Spider Leg

Sheet: /
File: Subsystem Schematic KG.kicad_sch

Title: Driver Nano 3 (Linear Actuator B + IMU) Subsystem Schematic

Size: A3	Date: 2026-10-21	**Rev: 1**
KiCad E.D.A. 10.0.7		Id: 1/1